



Oussama CHAABENE

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ABOUT ME

Full Stack developer with deep expertise in AI, LLMs, and distributed systems. Experienced in building RAG architectures, optimizing AI platforms, and delivering end to end applications. Seeking opportunities in Full Stack Development or AI Engineering.

WORK EXPERIENCE

 **ACCESS INTERNATIONAL** – TUNIS, TUNISIA

Website <https://www.linkedin.com/company/access-inter/>

AI SOFTWARE ENGINEER – 1 JUN 2025 – 30 JUN 2025

- **Developed** a multi-tenant reservation platform (Angular + Django) with a user interface for booking services (e.g., padel courts) and provider back-office dashboards for scheduling, insights, and full platform control (each provider on its own subdomain).
- **Built** a WhatsApp booking agent using n8n, integrating backend API calls to process reservations directly from chat.
- **Configured** SSH tunnelling + ngrok for secure webhook integration and SSL termination without certificate costs.

 **FAVIZONE** – TUNIS, TUNISIA

Website <https://favizone.com/>

AI ENGINEER – 1 MAY 2025 – 31 MAY 2025

- **Deployed and optimized** large language models (Mistral-8x7B, Llama-70B) with vLLM, leveraging GPU memory management and tensor parallelism.
- **Benchmarked performance** across hardware (H100, A10, A100, Tesla) for token throughput, latency, and concurrent users, optimizing VRAM usage and inference speed.
- **Integrated custom models** (e.g., JAIS family) with chat templates and tokenizer configurations for vLLM compatibility.
- **Implemented production monitoring** to track tokens/sec, memory usage, and real-time scaling metrics.
- **Enabled Ray clusters** on multi-node setups for distributed inference.

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AI SOFTWARE ENGINEER INTERN – 3 FEB 2025 – 31 MAY 2025

- **Built an end to end AI chatbot platform** that simplifies chatbot creation for non-technical users, turning a months-long process into an accessible workflow.
- **Implemented multi-tenant RAG architecture** with ChromaDB, semantic and row-based chunking, metadata filtering, and intelligent deduplication using LangChain.
- **Deployed multi language LLMs** (Mistral-Small-24B for English/French, Cohere Command-R7B for Arabic) with function calling on high-performance infrastructure (160GB VRAM).
- **Developed asynchronous data pipelines** with Celery, supporting multiple formats, web scraping via Selenium Grid, and database connectors.
- **Integrated automated prompt engineering** using COSTAR methodology and context optimization.
- **Designed production infrastructure** with containerized microservices, React frontend, and Django REST backend.
- **Built real-time analytics dashboards** for conversation tracking, CSV exports, and performance monitoring.

 **FREEDOM OF DEV** – SFAX, TUNISIA

Website <https://www.freedomofdev.com/en/homepage/>

AI ENGINEER INTERN – 1 JUL 2024 – 31 AUG 2024

- **Developed a ML model recommendation platform** where users upload Excel or ZIP files and select the target. the platform trains models using low-code (LazyPredict) and then Grid Searching for hyperparameter optimization and returns the top 3 models.
- **Built full-stack integration** with ReactJS frontend and Django backend for file uploads, model training, testing, and downloadable outputs.
- **Enabled users to download complete packages** including the trained model, scaler, and PDF guide.
- **Implemented image classification and custom model training** using YOLO and AutoKeras, with evaluation metrics and comparisons for easy understanding.

 **COMPITECHNOLGY** – SFAX, TUNISIA

MOBILE DEVELOPER INTERN – 1 JUL 2023 – 15 SEP 2023

- **Developed a mobile app prototype** for a Canadian startup using Flutter, aligned with business goals through Flutter and Figma UI designs.
- **Integrated Firebase** for backend functionality and real-time updates.

 **IBEFORUM** – REMOTE / BENGALURU, INDIA

DATA ENTRY INTERN – 1 JUN 2022 – 31 AUG 2022

- **Collected and curated data** using web-based tools and advanced search techniques to support organizational objectives.

● **EDUCATION AND TRAINING**

15 SEP 2022 – 2 JUL 2025 Sfax, Tunisia

ENGINEERING DEGREE Faculty Of Sciences Of Sfax

Website <https://fss.rnu.tn/useruploads/files/cycleingenieurinfo.pdf> | **Field of study** Data Engineering

15 SEP 2020 – 31 MAY 2022 Sfax, Tunisia

BACHELOR DEGREE (INTEGRATED PREPARATORY MATH, PHYSICS AND COMPUTER SCIENCE) Faculty Of Sciences Of Sfax

Website <https://fss.rnu.tn/useruploads/files/P2I-2021-2022.pdf>

15 SEP 2016 – 15 JUN 2020 Gabes, Tunisia

BACCALAUREATE Ibn Khaldun High School Mettouiia

Field of study Technology

● **LANGUAGE SKILLS**

Mother tongue(s): **ARABIC**

Other language(s): **ENGLISH** | **FRENCH**

● **SKILLS**

Languages and Frameworks

Python | Django | FastAPI | Flask | Java | Spring Boot | Javascript | React | Angular | C | SQL | IaC | Git | Linux | Docker | Kubernetes | Hadoop | Spark | Zookeeper | Hive | Kafka | Hbase | Hue | VLLM | Ollama | Redis | Celery | ChromaDB | RAG | LLM | N8N | LangChain

Software

VMware | VirtualBox | PyCharm | PowerBI | Visual Studio | Eclipse | Android Studio | DataSpell

● **PROJECTS**

MLOps Project

Built an MLOps pipeline for predictive modeling with data versioning, experiment tracking, monitoring, scalable deployment, and automated CI/CD, plus a React dashboard for insights.

Data Analytics Project

built a data pipeline using a dockerized Selenium Grid to scrape data from e-commerce sites (Tunisianet, Mytek, Zoom, Wiki), streamed it to Apache Kafka, stored it in Apache Hive, and analyzed it using Apache Hue for business insights.

Intelligent Waste Classification System

Developed a machine learning system to automatically classify organic vs. recyclable waste using 16,880 images, implementing both traditional ML approaches (SVM, Random Forest, XGBoost, KNN with hand crafted features) and deep learning models (CNN architectures like ResNet, VGG, and Transformers). Created an end to end pipeline capable of detecting, segmenting, and classifying multiple waste types in complex real world environments to address environmental pollution and improve recycling efficiency

MediReport AI

Developed a healthcare chatbot that acts as a generalist doctor, assisting users with medical inquiries and providing tailored advice. Built using a fine-tuned LLaMA model for accurate responses, a Flask API for seamless backend functionality, and a React-based interface for an intuitive user experience, the chatbot ensures clear, reliable communication to support better healthcare outcomes.

End Of Year Project

Developed a real-time data stream processing system for log analysis using Spark and Kafka, managed with YARN and deployed on Linux environments via VirtualBox, with analysis Using Pyspark and visualization facilitated through Chart.js.

Automated Infrastructure

Configured automation infrastructure with Kubernetes, managed repositories with Nexus, and implemented CI/CD with Jenkins. Used Vagrant, VirtualBox, Linux and GitLab

YouTube Sentiment Classifier

Classified Arabic comments into positive, negative, and neutral categories. Evaluated YouTube videos for pleasantness and ranked them using web scraping, NLP, NLTK, vectorizing, ML, and DL

ELK Log Monitoring System

Deployed and configured an ELK stack, incorporating Filebeat and Redis to enable real-time log collection, improving troubleshooting efficiency by 40% and streamlining incident response times across the operations team .

Pneumonia Detection

Detected pneumonia in X-ray images using deep learning. Built CNNs from scratch and used pretrained models like VGG16 and MobileNetV2 with freezing techniques

HONOURS AND AWARDS

2nd Place Local Solution Challenge

By Google

2nd Place PYFAC9 Competition

Focused on AI in the Automotive Field, organized by **PYFAC** and **Primatec**

Best IoT Project

Awarded by Technopole Sfax, IEEE Tunisian Section, and University of Sfax

Best Mentor at MASTER-FASTER

A Data Science Competition

● **VOLUNTEERING**

Member & Former Treasurer GDSC FSS

Former Treasurer & Member SecuriNets Club FSS

Former EfS Manager NATEG ENIS Student Chapter

Former Vice-Chair & Current Chair IEEE Computer Society FSS Student Branch Chapter.
